NC ALL FRC SAFE-ALL VALUE

PLANNER

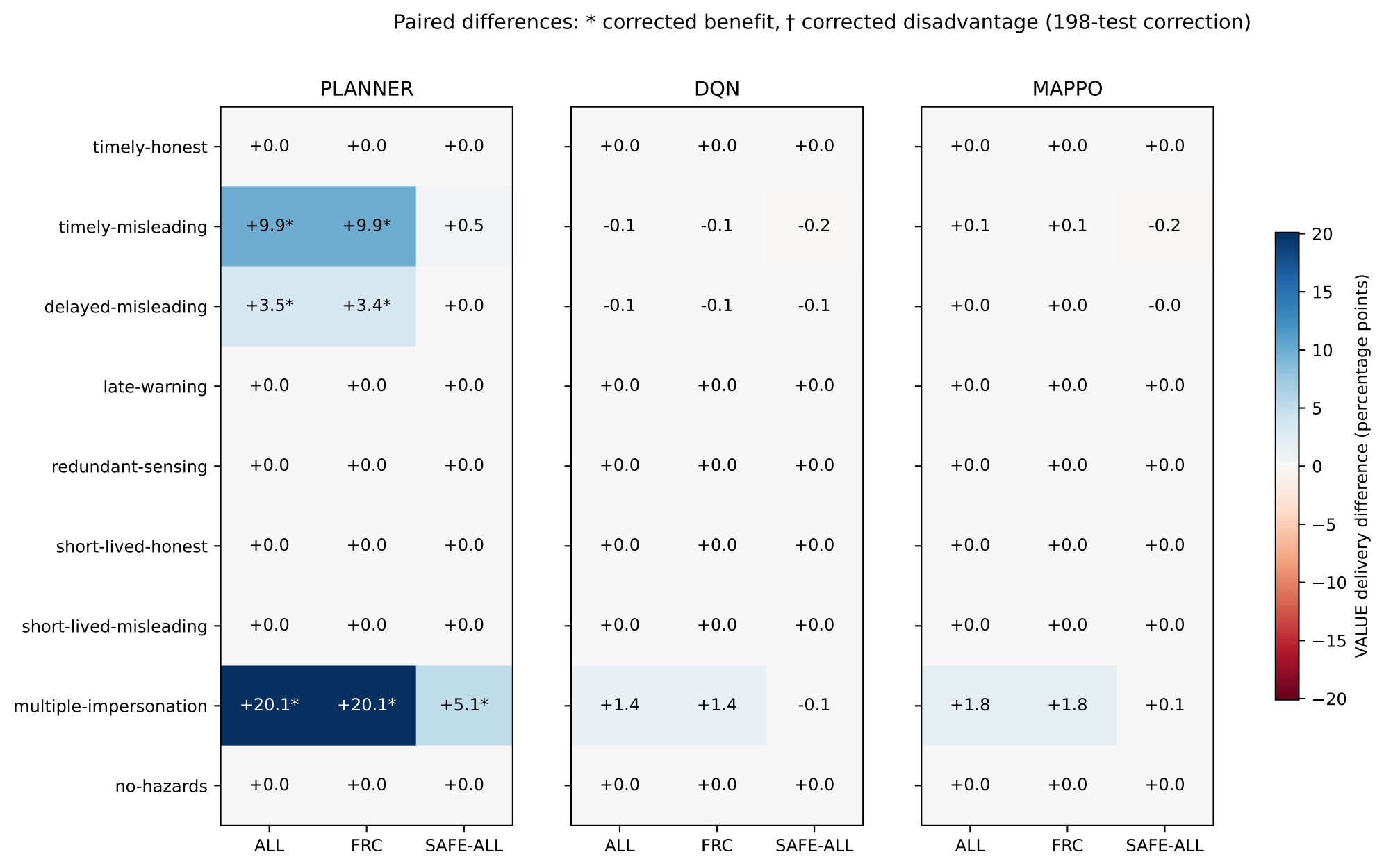
DQN

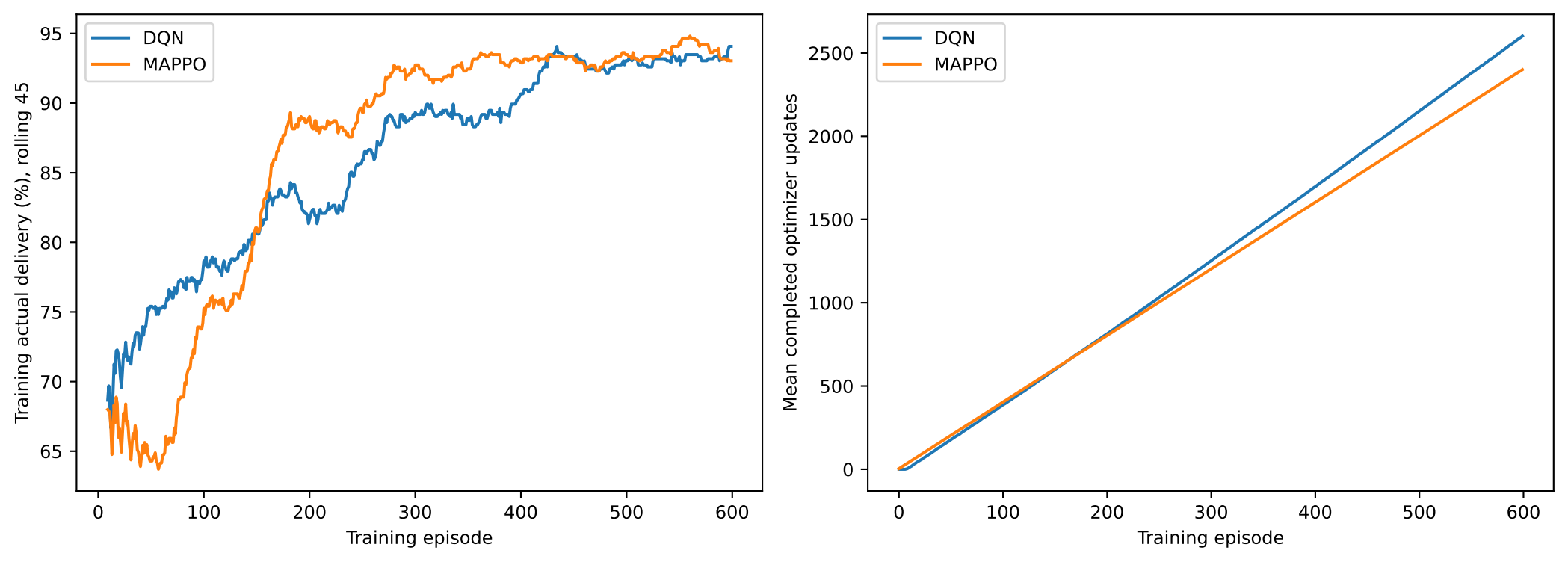
MAPPO

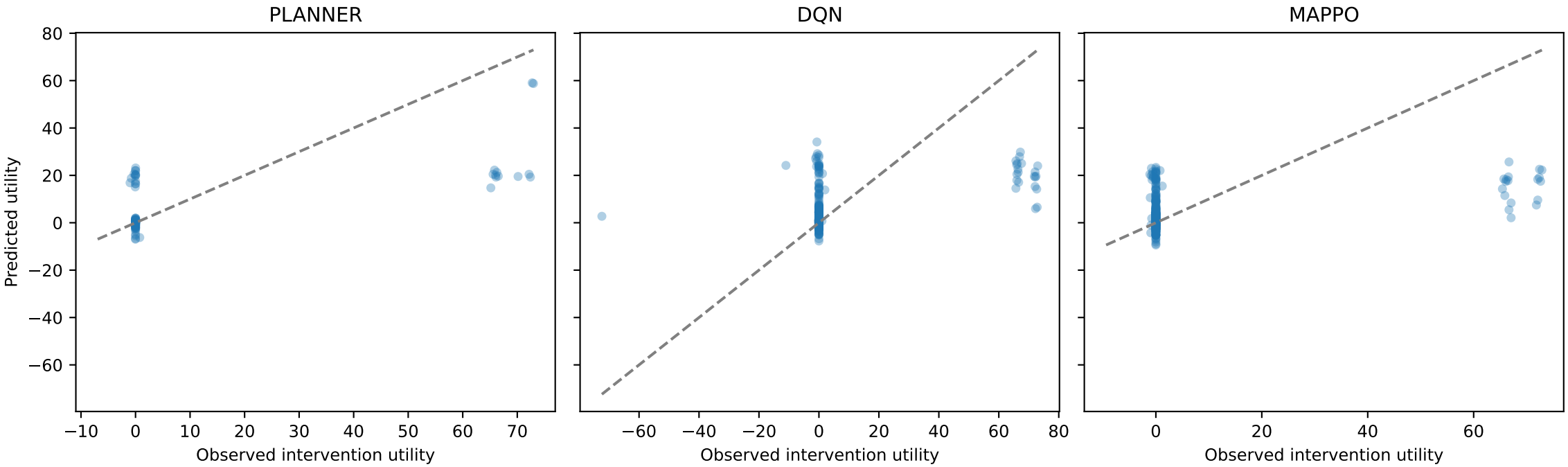
Actual delivery (%)

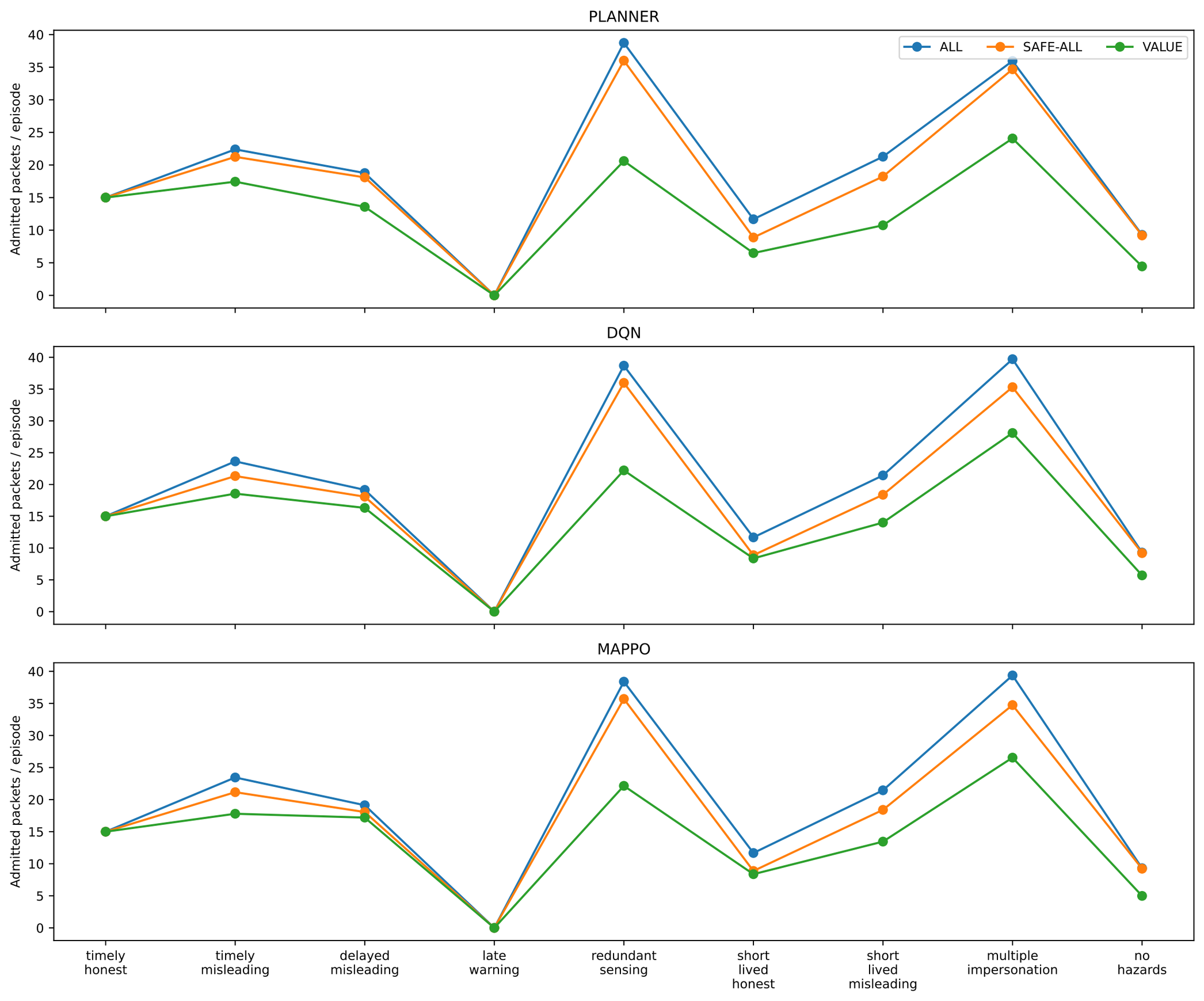
0 20 40 60 80 100

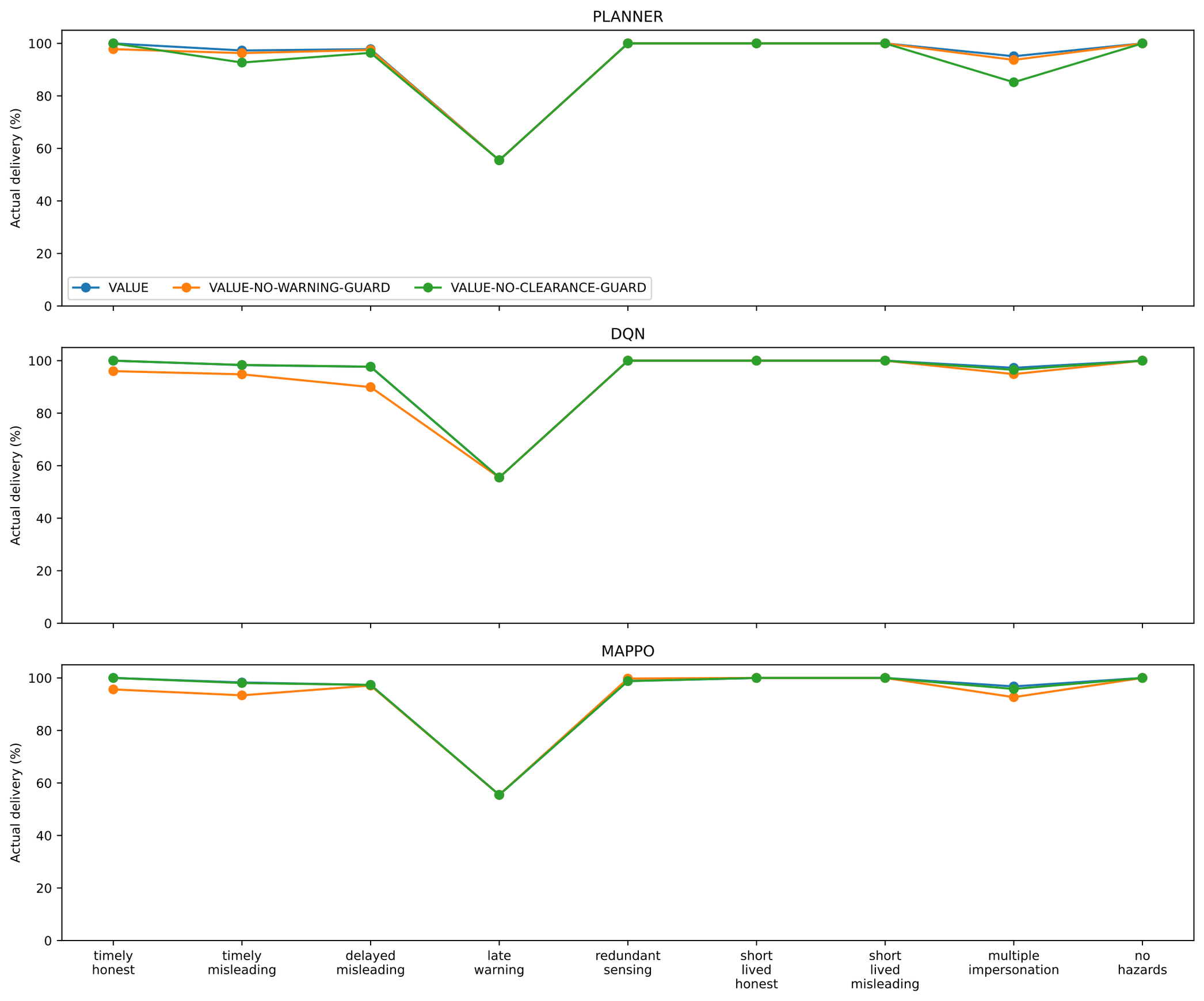
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards

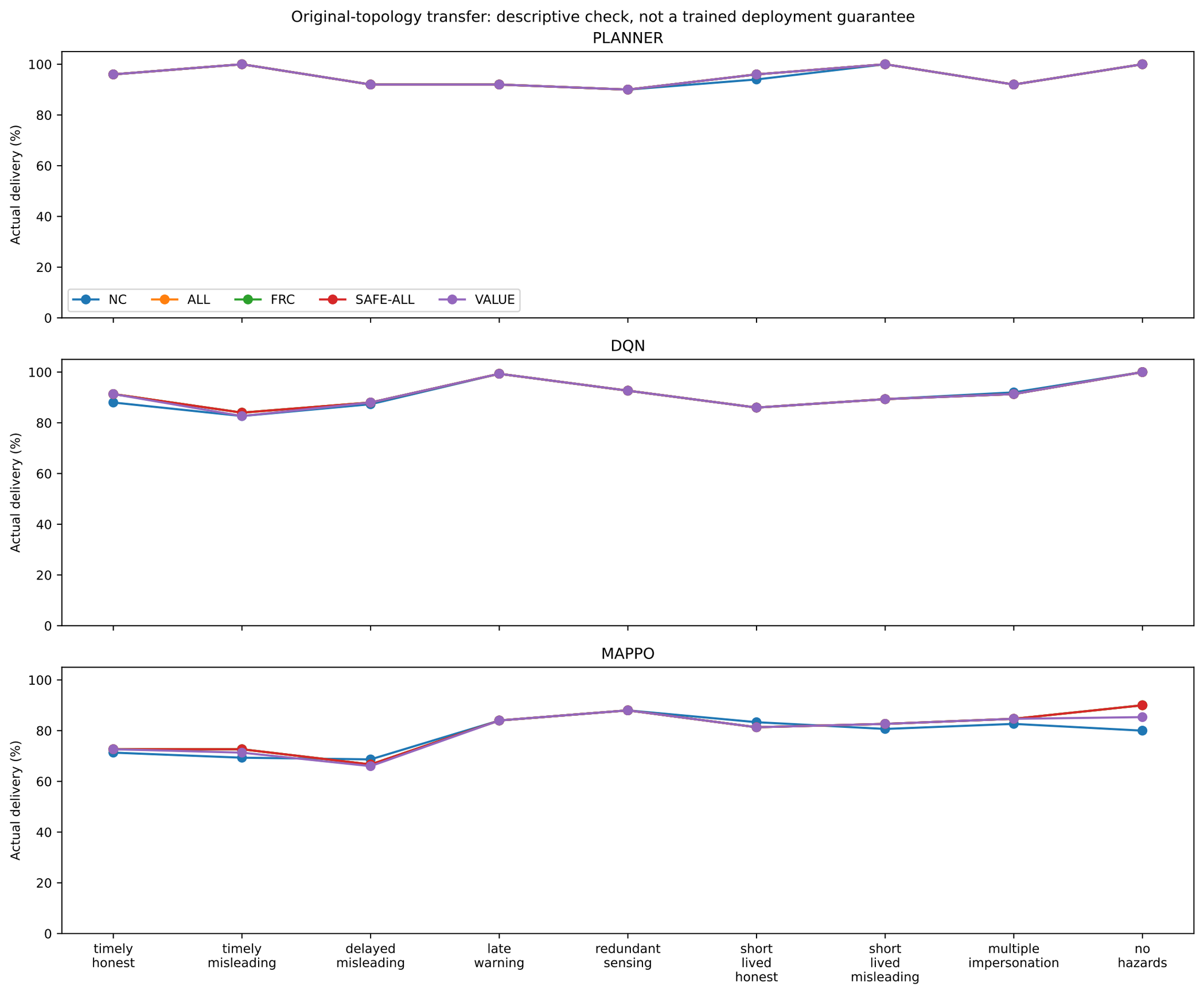

Paired differences: * corrected benefit, † corrected disadvantage (198-test correction)
PLANNER
DQN
MAPPO
timely-honest
timely-misleading
delayed-misleading
late-warning
redundant-sensing
short-lived-honest
short-lived-misleading
multiple-impersonation
no-hazards
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+9.9* +9.9* +0.5
-0.1 -0.1 -0.2
+0.1 +0.1 -0.2
+3.5* +3.4* +0.0
-0.1 -0.1 -0.1
+0.0 +0.0 -0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+20.1* +20.1* +5.1*
+1.4 +1.4 -0.1
+1.8 +1.8 +0.1
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
+0.0 +0.0 +0.0
ALL FRC SAFE-ALL
ALL FRC SAFE-ALL
ALL FRC SAFE-ALL
20
15
10
5
0
−5
−10
−15
−20
VALUE delivery difference (percentage points)

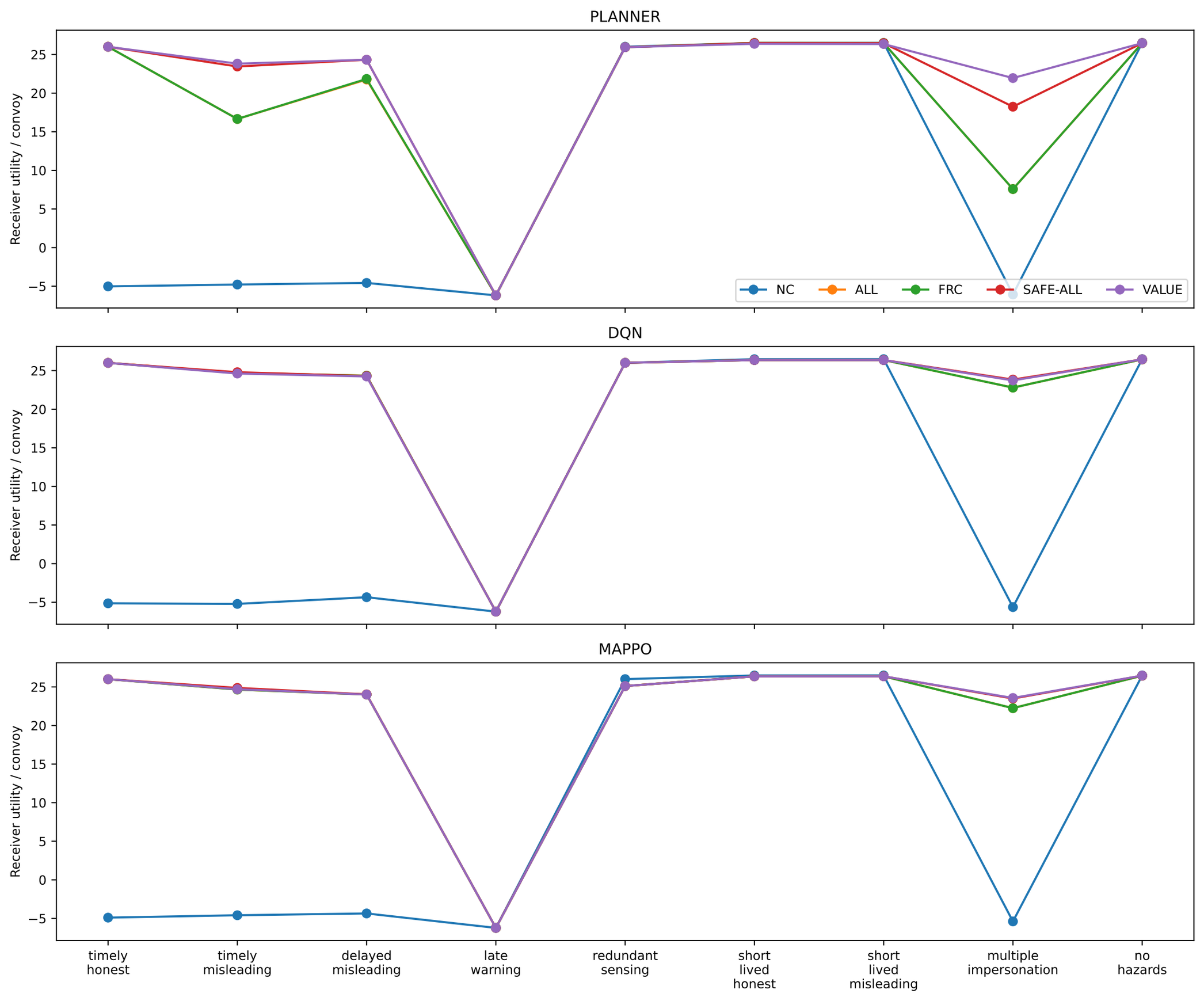

DQN
MAPPO
Training actual delivery (%), rolling 45
Training episode
Mean completed optimizer updates
Training episode


Internal held-out causal labels: episodes excluded from model fitting
PLANNER
DQN
MAPPO
Predicted utility
Observed intervention utility
Observed intervention utility
Observed intervention utility


PLANNER
DQN
MAPPO
ALL
SAFE-ALL
VALUE
Admitted packets / episode
0
5
10
15
20
25
30
35
40
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards


PLANNER
DQN
MAPPO
Actual delivery (%)
0
20
40
60
80
100
VALUE
VALUE-NO-WARNING-GUARD
VALUE-NO-CLEARANCE-GUARD
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards


Original-topology transfer: descriptive check, not a trained deployment guarantee
PLANNER
DQN
MAPPO
Actual delivery (%)
0
20
40
60
80
100
NC
ALL
FRC
SAFE-ALL
VALUE
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards


PLANNER
DQN
MAPPO
Receiver utility / convoy
NC
ALL
FRC
SAFE-ALL
VALUE
timely honest
timely misleading
delayed misleading
late warning
redundant sensing
short lived honest
short lived misleading
multiple impersonation
no hazards